

Sajal Kanwal

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EDUCATION

Lovely Professional University
B.Tech in Computer Science & Engineering (Data Science & ML) – CGPA: 8.68/10.0

Phagwara, Punjab
Aug’ 2023 – Present

TECHNICAL SKILLS

Languages	Python, SQL, TypeScript, JavaScript
AI & ML	PyTorch, TensorFlow, TorchScript, scikit-learn, RAG, LangChain
Data Engineering	Pandas, NumPy, SQLAlchemy, ETL/ELT, dbt, Apache Airflow, Jupyter
Databases	PostgreSQL, MySQL, Redis, Prisma, pgvector
Frontend	React, Next.js, Tailwind CSS, GSAP
Backend & APIs	Node.js, NestJS, FastAPI, REST APIs, tRPC, BullMQ
DevOps & Cloud	Docker, CI/CD, AWS (Rekognition, Lambda, DynamoDB, S3), Git, GitHub, Linux/Bash, Vercel

PROJECTS

Monocle: AI-Native GitHub Intelligence Platform

Aug’ 2026

Next.js 16 · tRPC v11 · Neon PostgreSQL (pgvector) · BullMQ/Redis · Google Gemini · Clerk · Stripe
<https://monocle-ai.vercel.app/>

- Architected an enterprise Zero-Waste RAG pipeline using Next.js 16, Neon PostgreSQL (**pgvector**), and Google Gemini, indexing 50+ repository AST files directly into 768-dim embeddings with 0 text-generation quota burn and <1s Time-To-First-Token (TTFT) streaming responses with exact file citations.
- Engineered an asynchronous distributed queue system using BullMQ and Upstash Redis with deterministic job deduplication and exponential backoff, eliminating 100% of serverless execution timeouts while orchestrating multi-tier model fallbacks (**gemini-2.5-flash** → **gemini-3.5-flash** → **gemini-flash-lite**) for 99.9% chat uptime.
- Built an end-to-end type-safe platform using tRPC v11, TypeScript, Clerk JWT auth, and Stripe Webhooks, implementing an idempotent credit ledger and a dual-mode editorial design system (Cafe/Espresso) with reactive cache synchronization across workspace dashboards.

Flux: Real-Time Market Microstructure Prediction Engine

Apr’ 2026

PyTorch · FastAPI · Binance WebSocket · Next.js 15 · TensorBoard · asyncio
<https://github.com/Sajal-kanwal/flux>

- Formulated a multi-task Transformer — VSN encoder, 2-layer LSTM, 4-head attention ($d_k=16$), dual-loss ($\alpha=0.6$ CE + Huber) — benchmarked across a 9-experiment ablation study against DeepLOB.
- Distilled 92 leakage-free microstructure features — OBI, trade flow, realised volatility — from 3 Binance WebSocket streams at 100ms cadence with rolling z-score normalisation and 70/15/15 temporal splits.
- Compressed inference via TorchScript + INT8 quantisation on FastAPI with a 100-step rolling buffer; piped live predictions via SSE into a Next.js 15 dashboard with LOB heatmap and attention overlays.

Vendor Performance Analysis: Retail ETL & Analytics Pipeline

Apr’ 2026

Python · MySQL · SQLAlchemy · Pandas · pytest
<https://github.com/Sajal-kanwal/Vendor-Performance-Analysis>

- Built a batch ETL pipeline that loaded 15.6M+ records across 6 raw retail CSV sources (sales, purchases, inventory, vendor invoices) into MySQL with connection pooling and batched inserts, achieving a 100% load success rate.
- Wrote CTE-based analytical SQL joining 6 tables to generate a vendor performance summary with 20+ derived business metrics, then engineered 17+ ML-ready features across 7 business-driven feature layers.
- Designed a profile-based data quality framework (schema, null, duplicate, range, and z-score outlier checks) and a multi-source sync pipeline (API/FTP/SFTP/DB) with scheduled automation and failure alerting.

CERTIFICATIONS

Agentic AI Certified Foundations Associate — Oracle	Jul’ 2026
Oracle Cloud Infrastructure 2025 Certified Generative AI Professional — Oracle	Oct’ 2025
Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate — Oracle	Oct’ 2025

ACHIEVEMENTS

- Top 10 — AWS College Hackathon
Developed a facial recognition attendance system using AWS Rekognition, Lambda, and DynamoDB, automating roll-call for 500+ students with 99.2% accuracy and sub-3s real-time processing.

Feb’ 2025